



Development and assessment of a complete-detoxication strategy for *Fuzi* (lateral root of *Aconitum carmichaeli*) and its application in rheumatoid arthritis therapy

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ARTICLE INFO

Article history:

Received 28 July 2012

Received in revised form

3 December 2012

Accepted 21 January 2013

Available online 29 January 2013

Keywords:

Detoxication

Decoction

Fuzi

Acute toxicity

Aconitine

Adjuvant arthritis

ABSTRACT

Ethnopharmacological relevance: *Fuzi* (lateral root of *Aconitum carmichaeli*) is a popular traditional Chinese medicine well known for its both therapeutic and high-toxic activities. Its toxic alkaloid ingredients, mainly aconitine, mesaconitine, and hypaconitine, are responsible for the high toxicity. However, to date, no detoxication strategy is available to completely eliminate *Fuzi*'s toxicity, and, whether *Fuzi*'s efficacy could be kept after detoxication, remain unknown and debatable.

Materials and methods: The purpose of this study was to establish and validate a complete-detoxication strategy for *Fuzi* via acute toxicity test, to clarify the detoxication mechanism by HPLC and titrimetric analyses, and to evaluate the therapeutic effect of detoxicated *Fuzi* on adjuvant arthritis (AA). Three processed *Fuzi* (*Bai-fu-pian*) with 30-min, 60-min, and 120-min decoctions, respectively, named *dBfp*-30, *dBfp*-60, and *dBfp*-120, were prepared for this study. For the acute toxicity test, their oral doses to male and female Kunming mice were up to 70–190 g/kg body weight, and their toxicological profiles were evaluated by median lethal dose (LD_{50}), maximal tolerance dose (MTD), minimal lethal dose (MLD), no-observed-adverse-effect-level (NOAEL), and time-concentration-mortality (TCM) modeling methods using a 14-day schedule with up to five doses. The HPLC analysis was performed to determine the detoxication-induced changes in composition and amount of aconitine, mesaconitine and hypaconitine in *Fuzi*, whilst the titrimetric method was adopted to estimate the amount changes of *Fuzi*'s total alkaloids. AA model was established by incomplete Freund's adjuvant injection in Wistar rats, and the animal's physiological (body weight, food intake, etc.), clinical (hind paw volume), and immunological (IL-1 and TNF- α) parameters were assessed as markers of inflammation and arthritis.

Results: With increasing decoction time, the acute toxicity of detoxicated *Fuzi* became decreased in the following order: *dBfp*-30 (LD_{50} of 145.1 g/kg; MTD of 70 g/kg; MLD of 100 g/kg; NOAEL of 70 g/kg) > *dBfp*-60 (too large LD_{50} ; MTD of 160 g/kg; MLD of 190 g/kg; NOAEL of 100 g/kg) > *dBfp*-120 (no LD_{50} ; unlimited MTD; unlimited MLD; NOAEL of 130 g/kg). *dBfp*-30 and *dBfp*-60 displayed the toxicity at a dose-dependent manner with maximum mortalities reaching 100% and 50% respectively, whereas no mortality or signs of intoxication was induced by *dBfp*-120. The chemical analyses revealed a dramatic reduction of the toxic alkaloids as well as total alkaloids in *Fuzi* after the detoxication, from which no level of aconitine and only minimum residual of mesaconitine ($0.56 \pm 0.02 \mu\text{g/g}$) and hypaconitine ($8.73 \pm 0.13 \mu\text{g/g}$) were detected in *dBfp*-120. However, no significant difference of total alkaloid amount was found among *dBfp*-30, *dBfp*-60, and *dBfp*-120 ($P > 0.05$), suggesting an equivalent conversion from toxic alkaloids to its non-toxic derivants in *dBfp*-120. Further, also no significant differences were seen among *dBfp*-30, *dBfp*-60, and *dBfp*-120 for the therapeutic effects on physiological, clinical, and immunological parameters in AA rat, indicating that *dBfp*-120 is of non-toxicity and efficacy.

Conclusions: A complete-detoxication strategy has been developed successfully for ensuring the safe and effective use of *Fuzi*. The detoxication mechanism associated with elimination of toxic alkaloids has kept

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Fuzi's efficacy, indicating a non-interdependent relationship between its efficacy and toxicity. This is the first report on such an optimal detoxication strategy and on the application of detoxicated *Fuzi* in AA. It may provide in depth understanding to the toxicological and pharmacological profiles of *Fuzi* and further benefit the herbal drug development with safety and efficacy for disease especially RA therapy.

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1. Introduction

Rheumatoid arthritis (RA) is a common chronic autoimmune and inflammatory disease affecting multiple joints, which progressively leads to joint destruction, disability, systemic complications, and even premature death in patients (Firestein, 2003). Joint (synovium) and systemic inflammation with hyperplasia (swelling) develops during RA process by which abnormal expression of pro-inflammatory cytokines such as interleukin 1 (IL-1) and tumor necrosis factor- α (TNF- α) have been triggered at inflammation sites (Zwerina et al., 2005; McInnes and Schett, 2011), resulting in damage of cartilage, bone, and tendons as well as life-threatening extra-articular complications (Scott, 2011). Currently, anti-inflammatory therapy using antirheumatics remains the major recommended strategy against RA (Campbell, 1998; Scott et al., 1998). However, most of those drugs have unsatisfactory efficacy as well as side effects (such as gastrointestinal disorders and renal morbidity) especially when treated with high doses and/or long-term administration (Pincus et al., 1992; Vane et al., 1994; Rainsford, 1999). Accordingly, a new therapeutic candidate with more efficacy and safety is indeed in demand for RA treatment.

Fuzi, the lateral root of *Aconitum carmichaeli* Debx (family Ranunculaceae), is originally produced and widely distributed over the southwest of China. As one of the most important traditional Chinese medicine (TCM), *Fuzi* has been extensively used as cardiogenic, analgesic, anti-inflammatory, and diuretic agents to treat colds, polyarthralgia, diarrhea, heart failure, beriberi, and edema for thousands of years (Murayama et al., 1991). Originally in the Eastern Han Dynasty of China (24–220 AD), *Fuzi* was recorded by 'Shennong Materia Medica' (Shennong Ben Cao Jing), the earliest Pharmacopeia of China, and was described for its medicinal effect against chronic arthritic diseases by 'Treatise on Febrile Diseases' (Shang Han Lun), a classical TCM book (Zhang et al., 1999). Its anti-rheumatism and anti-arthralgia effects exerted through dispelling the wind-cold-damp evil have been presented in the Chinese Pharmacopeia (China Pharmacopeia Committee, 2010). Modern clinical studies have evidenced that such effects of *Fuzi* in combination with other herbs have anti-RA property in the therapy of patients (Huang et al., 2010), indicating its promising therapeutic potential against RA. However, paradoxically, *Fuzi* is well-known also for its evil

reputation as an extremely toxic plant in which toxic aconite alkaloids are contained (Chan, 2009).

Aconite alkaloids in *Fuzi* can be divided into C₁₉- and C₂₀-diterpenoid alkaloids according to their skeleton structures (Hu et al., 2009). The C₁₉-diterpenoid alkaloids share a common C₁₉-norditerpenoid skeleton and display four main types based on the substituents at C₈ and C₁₄, of which diester-diterpenoid alkaloids (DDAs) with acetyl group at C₈ and benzoyl ester group at C₁₄ are the toxic alkaloids (Ameri, 1998a). Aconitine, mesaconitine and hypaconitine, the main DDAs (Fig. 1), serve as highly toxic cardiotoxins and neurotoxins due to their hyperpolarization and activation effect on the voltage-dependent sodium channels of the myocardium, nerves, and muscles cell membranes (Ameri, 1998a; Chan, 2009), resulting in a toxidrome consisting of neurological, cardiovascular, and gastrointestinal signs and symptoms (Lin et al., 2004; Chan, 2009). Indeed, the amounts and composition of those aconitine-type compounds can determine the toxicity of DDA-rich herb such as *Fuzi*. Thus, a pretreatment processing by which to reduce such compounds is absolutely essential for the safe application of *Fuzi* in clinic. Traditionally, soaking, saturating, steaming, and boiling preparation has been designed for *Fuzi* processing, in which the three DDAs could be hydrolyzed into non-toxic derivatives (non-ester diterpenoid alkaloids) by losing their acetyl and benzoyl ester groups at C₈ and C₁₄ respectively (Fig. 1), resulting in toxicity reduction (Chan et al., 1994; Huang et al., 2007). After the traditional processing with modifications, various processed *Fuzi* – *Bai-fu-pian*, *Hei-shun-pian*, *Yan-fu-zi* – are available with toxicity less than raw *Fuzi* (Liou et al., 2005; China Pharmacopeia Committee, 2010). *Bai-fu-pian* and *Hei-shun-pian* have been reported to contain the relatively lower amounts of toxic alkaloids than other processed *Fuzi* types, indicating a much safer nature of them (Lu et al., 2010). Nonetheless, two problems still exist for those processed *Fuzi*: first, poisoning often occurs after intake of the processed *Fuzi* since its toxic alkaloids remain present at hazardous levels, owing to the inadequacy of processing as well as the faulty use of high dose (Tai et al., 1992; Chan et al., 1994); second, *Fuzi's* therapeutic activity (anti-inflammatory, analgesic, and cardiogenic activity) seems to have relevance to the presence of toxic alkaloids (Ameri, 1998a; Bello-Ramírez et al., 2003), suggesting a possibility that *Fuzi* may be ineffective

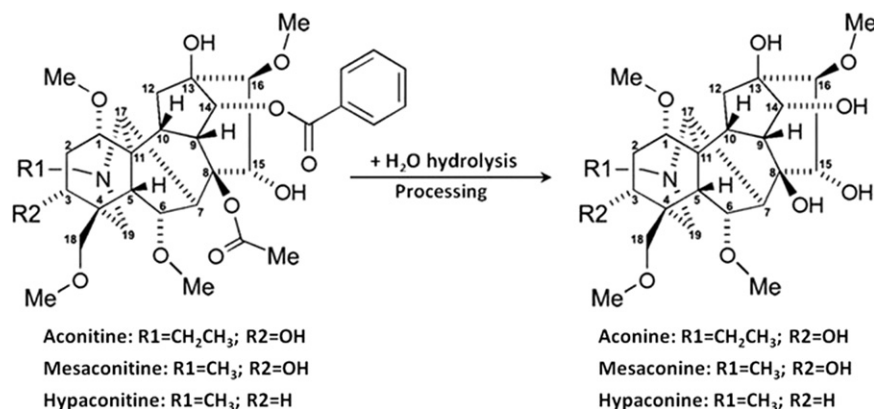


Fig. 1. Chemical structures of major aconite alkaloids and the reaction from toxic aconitines to nontoxic aconines. Me=CH₃ (methyl group).

without its toxicity. It is not possible to achieve the safe and effective application of *Fuzi* in clinic, unless those problems can be addressed. For this, we proposed a hypothesis that the toxicity is not necessary for the efficacy of *Fuzi* and then tested it in this study.

Our preliminary study has discovered that *Bai-fu-pian* (*Bfp*) contained the lowest amounts of toxic alkaloids among others (data not shown). Therefore, in this study, we adopted *Bfp* to conduct following experiments for testing our hypothesis: (a) development and optimization of a detoxication strategy for *Fuzi* using acute toxicity test; (b) Chemical analyses of *Fuzi*'s toxic alkaloids and total alkaloids for explaining the detoxication mechanism; and (c) evaluation of the efficacy of detoxicated *Fuzi* in rat adjuvant arthritis (AA) model. During this study, a standardized optimal detoxication strategy for *Fuzi* may be created and applied in arthritis treatment for the first time.

2. Materials and methods

2.1. Plant materials and *Bfp* preparation

The daughter root of *Aconitum carmichaeli* Debx (family Ranunculaceae) was harvested from its trueborn cultivating area in Jiangyou (Sichuan Province, China), and was authenticated by the authors. A voucher specimen (No.: 081101) has been deposited at Institute of Orthopaedics and Traumatology, Zhejiang Chinese Medical University, China. Its raw material was directly cut into slices and dried without any processing, named raw *Fuzi*.

Bfp was prepared through a traditional processing proposed by the Chinese Pharmacopeia (China Pharmacopeia Committee, 2010). Firstly, the raw root was washed in water and soaked in edible mother liquor of mineral salt for several days, followed by thorough boiling in the soaking liquid and rinsed with water. Then it was peeled and longitudinally sectioned into approx. 3 mm slice thickness and was again soaked and rinsed in water. After being thoroughly steamed and completely dried by the sun, *Bfp* was obtained with white appearance. Its standard product (Batch no.: 100424) was purchased from Zhejiang Chinese Medical University Medicine Yinpian Factory (Hangzhou, China).

2.2. Chemicals and reagents

The reference standards of aconitine (Batch no.: 0720-9406), mesaconitine (Batch no.: 0799-9203), and hyaconitine (Batch no.: 0798-9202) were supplied by the National Institute for the Control of Pharmaceutical and Biological Products (Beijing, China). The purities of the three chemicals were above 99.8%. Acetonitrile (ACN) and tetrahydrofuran (THF) were of HPLC grade (Tedia, Fairfield, USA). The distilled water was prepared and purified by Milli-Q system (Millipore, Milford, MA, USA). Other commercial reagents (isopropanol, methanol, triethylamine, ethyl ether, ethyl acetate, dichloromethane, aqueous ammonia, ammonium acetate, and acetic acid) were of analytical grade.

2.3. Animals

Male Wistar rats weighing 180–220 g and Kunming mice with both sexes weighing 18–22 g were purchased from the Shanghai Laboratory Animal Center of Chinese Academy of Sciences (SLAC-CAS, Shanghai, China; Grade SPF II Certificate no. SCXK2007-0005). The animals were kept in an environmentally controlled breeding room (temperature of $22 \pm 1^\circ\text{C}$, relative humidity of $60 \pm 10\%$, and a 12/12 h light/dark cycle) for at least 1-week acclimatization. All animals were fed rodent laboratory chow with tap water ad libitum during the testing periods. All

experimental procedures were in strict accordance with the China legislation on the use and care of laboratory animals and with the guidelines established by the Institutional Animal Ethics Committee and Committee for the Purpose of Control and Supervision of Experiments on Animals in China.

2.4. Detoxication treatment

Bfp was powdered and soaked in 5-fold stilled water for 30 min. The mixture was equally separated into three samples and decocted for 30 min, 60 min, and 120 min, respectively, following boiling. The first-stage extract of each sample was obtained after filtration. The residue was again decocted with 4-fold water for 30 min, 60 min, and 120 min, respectively, to obtain the second-stage extract. Finally, the first-stage and second-stage extracts of each sample were mixed together and centrifuged at 3000 rpm for 5 min. Each supernatant was collected, filtered and concentrated to dryness and named 30-min decocted *Bfp* (dBfp-30), 60-min decocted *Bfp* (dBfp-60), and 120-min decocted *Bfp* (dBfp-120), respectively. An aliquot of them was mixed with aqueous ammonia and vortexed for 2 min, followed by mixing with 5-fold ethyl ether overnight at 4°C . The ethyl ether layer was collected and taken to dryness, and the residue was dissolved in hydrochloric acid-methanol for the HPLC analysis. Another aliquot was dissolved in distilled water for oral administration in animal experiments. The decocted *Bfp* were designated as the detoxicated *Fuzi*.

2.5. HPLC quantitative analysis

2.5.1. Instruments and conditions

The HPLC analysis was performed on a LUMTECH HPLC system (Knauer, Berlin, Germany) equipped with K-2501 UV detector. Chromatographic separation was achieved on a Hypersil BDS-C₁₈ column (5 μm , 210×4.6 mm ID) coupled with Hypersil BDS-C₁₈ pre-column (5 μm , 10×4.6 mm² ID) (Shandon Scientific, Cheshire, UK) at temperature of 30°C . The mobile phase consisted of acetonitrile and 0.1 triethylamine (45:55, v/v) with flow rate of 1.0 ml/min. The sample injection volume was 2 μl for raw *Fuzi* and 20 μl for others. The detection wavelength was 235 nm. The data were analyzed using ChromGate software (Knauer, Berlin, Germany). Two-point external standard method was used for the quantitative determination of aconitine, mesaconitine, and hyaconitine following three parallel analyses.

2.5.2. Preparation of standard solutions

The reference standards of aconitine, mesaconitine, and hyaconitine were dissolved in isopropanol–dichloromethane (1:1, v/v) solution at concentrations of 166.7 $\mu\text{g/ml}$, 259.8 $\mu\text{g/ml}$, and 682.3 $\mu\text{g/ml}$, respectively. These stock solutions were diluted to obtain working standard solutions of aconitine (1.04–33.35 $\mu\text{g/ml}$), mesaconitine (1.63–51.96 $\mu\text{g/ml}$), and hyaconitine (14.28–136.46 $\mu\text{g/ml}$).

2.5.3. Method validation

The validation of current HPLC method was determined for aconitine, mesaconitine, and hyaconitine. Intra-day and inter-day precisions were determined in 1 day and over 5 days, respectively, shown as relative standard deviation (RSD). Repeatability was evaluated by determining six replicates of each sample and expressed as RSD. Recovery was calculated by comparing measured concentrations of the analytes to the nominal values and expressed as mean values with RSD. Stability was evaluated by analyzing *Fuzi* sample at 0, 4, 8, 12, 24 and 48 h during its routine storage, and was expressed as RSD by comparing the stored sample with fresh one.

2.6. Determination of total alkaloid content

Total alkaloid content of raw *Fuzi*, *Bfp*, and detoxicated *Fuzi* was determined using a titrimetric method recommended by the Chinese Pharmacopeia (China Pharmacopeia Committee, 2010) with some modifications. Briefly, each vacuum-dried sample was mixed with ethyl ether-chloroform (3:1, v/v) solution added with aqueous ammonia solution, and then was sealed overnight. After filtration, the filtrate was collected and freeze-dried, followed by dissolving in ethanol. For titration, the mixture was added to a known volume of sulfuric acid titrant and titrated with sodium hydroxide titrant, using methyl red as indicator. Content of total alkaloid was calculated in terms of aconitine which concentration was determined by the volume of sulfuric acid titrant (1:12.9, v/w). This method was replicated for three times.

2.7. Acute toxicity test

Equal amount of male and female Kunming mice were randomly divided into 16 groups with 16 animals each. Before oral administration, all mice were fasted for 12 h with water ad libitum. Each five groups were orally given one type of detoxicated *Fuzi* (dBfp-30, dBfp-60 or dBfp-120) in doses of 70, 100, 130, 160, and 190 g/kg, respectively. The control group received an equal volume of water. After a single administration, the animals were observed closely during first 24 h and were kept under observation up to 14 days. Their toxic signs and manifestations, such as abnormal motor activity, alteration in food and water intake, and diarrhoea, were monitored and recorded. The body weight change of each mouse was measured, and the survival rates calculated. The necropsy of dead mice was performed to observe the gross pathological changes of the major organs and tissues. Acute toxicity of each detoxicated *Fuzi* was evaluated using median lethal dose (LD₅₀), maximal tolerance dose (MTD), minimal lethal dose (MLD), and no-observed-adverse-effect-level (NOAEL) methods. Time–concentration–mortality (TCM) modeling method was also employed for the toxicity analysis.

2.8. Therapeutic evaluation of the detoxicated *Fuzi* in AA model

2.8.1. Experimental design

The 110 rats were divided randomly and equally into 11 groups. Each rat except that in normal group were single-injected with 0.1 ml incomplete Freund's adjuvant (Sigma, St. Louis, USA) added by 10 mg/ml heat-inactivated *Mycobacterium butyricum* (Sigma, St. Louis, USA) intradermally into the right hind metatarsal footpad to establish the AA model. Then each treated group received oral administration of dBfp-30, dBfp-60, and dBfp-120 at 14 g/kg, respectively, for next 15 days. The normal group was given an equal volume of distilled water. During this period, physiological and clinical parameters were measured. At the end, all rats were sacrificed under anesthesia and their blood sample was immediately taken for immunological analysis.

2.8.2. Physiological measurement

The hair, feces and behavior pattern were daily observed for each rat while the body weight, rectal temperature, food and drink intake, urine volume were measured every 3 days during the experimental period.

2.8.3. Clinical evaluation

Hind paw volume (HPV), the basic clinical parameters of arthritis, was monitored every 2 or 3 days till the end of the experiment. HPV was measured by an electronic water plethysmometer (Ugo Basile, Milan, Italy) and expressed as a percentage

increase in terms of the equation $HPV\% = (V_t - V_n)/V_n$, where V_n represents the paw volume before injection and V_t the paw volume after injection.

2.8.4. Immunological analysis

Each rat blood sample was obtained from the abdominal vein and centrifuged (3000 rpm, 10 min, 4 °C) for serum ELISA analysis. Quantitative detection of rat IL-1 and TNF- α was conducted by an enzyme immunoassay analyzer (model 680; Bio-Rad, Hercules, USA) using commercial ELISA kits (Hyperheal Biotech, Shanghai, China). The test was done according to the manufacturer's instructions.

2.9. Statistical analysis

Considering the possible effect of both time and dose variables on the toxicity of detoxicated *Fuzi*, a TCM modeling method (Feng et al., 1998) was applied to analyze the resulting data from acute toxicity test. The mice mortality (q_{ij}) caused by the given dose (d_i) of detoxicated *Fuzi* at specific time intervals (0, 24, 48, 72, 120, 168, 336 h) after administration were corrected with background mortality observed from the control group and then fitted to the TCM model

$$q_{ij} = 1 - \exp[-\exp(\gamma_j + \beta \log_{10}(d_i))]$$

by a maximum likelihood approach (Nowierski et al., 1996). The estimates β and γ_j describe the dose effect and the conditional time effect of d_i , respectively. The goodness for each fit was tested with Hosmer–Lemeshow statistic C, a grouped Pearson's χ^2 . The fitted β and γ_j estimates were then used to yield the cumulative TCM model

$$p_{ij} = 1 - \exp[-\exp(\tau_j + \beta \log_{10}(d_i))]$$

where p_{ij} is the cumulative mortality and $\tau_j = \ln(\sum_{k=1}^j e^{\gamma_k})$ the cumulative time effect. As an index for acute toxicity, LD₅₀ with associated 95% confidence limits (CL) was determined by the Bliss's method. Other measurements were expressed as the mean $\pm$ standard deviation and subjected to one-way analysis of variance (ANOVA), followed by Fisher's least significant difference (LSD) comparison. P value of <0.05 was considered statistically significant. All analyses were performed using an updated version of DPS software (Tang and Feng, 2007).

3. Results

3.1. Method validation of HPLC assay

The precision of this method, evaluated by performing five replicate assessments, was acceptable with both intra-day and inter-day RSD of less than 2% (Table 1). The repeatability of this assay was good, due to the peak area RSD of each compound in *Fuzi* extract being less than 3% (Table 1). The recovery of these compounds ranged from 96.77 to 105.45% with RSDs less than 3%, indicating its acceptability under this method condition (Table 1).

Table 1
Precision, repeatability, recovery, and stability evaluation for the HPLC assay.

Samples	Precision (RSD, %)		Repeatability (RSD, %)	Recovery		Stability (RSD, %)
	Intra-day	Inter-day		Mean (%)	RSD (%)	
Aconitine	1.72	0.75	1.70	98.90	1.49	2.79
Mesaconitine	1.83	0.82	2.80	99.93	2.94	2.74
Hypaconitine	1.60	1.30	1.95	102.72	1.81	1.87

A good stability of these compounds in *Fuzi* was also observed for at least 48 h at room temperature, since each peak area RSD was less than 3% (Table 1).

3.2. HPLC analysis of aconitine-type compounds in different *Fuzi* types

The HPLC chromatographic comparison of each aconitine-type compound among raw *Fuzi*, *Bfp*, and detoxicated *Fuzi* were shown in Fig. 2. The retention time of aconitine, mesaconitine, and hyaenitine in all samples were 16.679–17.139 min, 9.400–9.949 min, and 35.719–36.649 min, respectively. The results

showed no remarkable chromatographic differences between raw *Fuzi* and *Bfp*. Nevertheless, after decoction treatment, a gradual reduction of chromatographic peak height was found in a time-dependent manner for each compound, and the peak of aconitine was even found disappeared after 120 min decoction (Fig. 2F).

Fig. 3 illustrated the concentrations of the three compounds in raw *Fuzi*, *Bfp*, and detoxicated *Fuzi*. The highest concentrations, which attained 405.7 ± 10.5 $\mu\text{g/g}$ for aconitine, 2660.1 ± 19.0 $\mu\text{g/g}$ for mesaconitine, and 3156.6 ± 16.4 $\mu\text{g/g}$ for hyaenitine, were seen in raw *Fuzi* and was far in excess in comparison with that of other *Fuzi* types. In *Bfp*, the concentration of each compound decreased significantly but remained much higher than that in detoxicated *Fuzi*. Interestingly, a stepwise concentration decrease of each compound was observed in detoxicated *Fuzi* with increasing decoction times. In particular, since the time reached 120 min, the three compounds in *dBfp*-120 displayed the lowest levels of 'none' for aconitine (undetectable), 0.56 ± 0.02 $\mu\text{g/g}$ for mesaconitine, and 8.73 ± 0.13 $\mu\text{g/g}$ for hyaenitine. This result was consistent with the above finding obtained from chromatographic analysis (Fig. 2).

3.3. Total alkaloid measurement of different *Fuzi* types

The amounts of total alkaloids in raw *Fuzi*, *Bfp*, *dBfp*-30, *dBfp*-60, and *dBfp*-120 were measured and plotted in Fig. 4, which present a strikingly significant difference among those samples ($P < 0.001$). Raw *Fuzi* with no processing treatment contained the maximum amount of total alkaloids which was significantly higher than that in the processed *Fuzi*, *Bfp* ($P < 0.001$). After decoction, total alkaloid amount reduced significantly from 1.80 ± 0.04 mg/g in *Bfp* to 0.97 ± 0.02 mg/g in *dBfp*-30 ($P < 0.001$). However, no matter how long time the decoction was maintained, there was no significant difference of total alkaloid amount among *dBfp*-30, *dBfp*-60, and *dBfp*-120 ($P > 0.05$), suggesting no influence of decoction time factor on the total alkaloids.

3.4. Acute toxicity of decocted *Bfp* types

Our preliminary experiment found the optimal dose range of detoxicated *Fuzi* being 70–190 g/kg for the acute toxicity test, since higher dose could not be tested due to experimental constraints in dissolving the extract of each sample and in its oral administration. During this 2-week period, only *dBfp*-30 and *dBfp*-60 could cause death of mice with both sexes (Table 2). Within 24 h, the death was initiated by *dBfp*-30 at 130 g/kg and *dBfp*-60 at 190 g/kg with mortalities of 12.5% and 25%, respectively (Fig. 5). Only *dBfp*-60 at 190 g/kg was lethal, while *dBfp*-30 at 100 g/kg could cause death within 72 h, indicating these lethal doses (100 g/kg and 190 g/kg) as the MLDs for *dBfp*-30 and *dBfp*-60, respectively. At the end, the mice exposed to *dBfp*-30 were all dead and the data fitted well to the TCM model with an acceptable goodness of fit ($P > 0.05$ for the Hosmer–Lemeshow test), indicating the toxicity of *dBfp*-30 being of both time- and dose-dependent manner. *dBfp*-60 showed lesser acute toxicity than *dBfp*-30 (Fig. 5). Considering the non-toxicity of *dBfp*-30 at 70 g/kg and *dBfp*-60 at 160 g/kg, these two doses could be estimated as their MTDs, respectively. However, *dBfp*-120 caused neither morbidity nor death with no LD₅₀, and unlimited MTD and MLD over such high dose range, indicating its certain safety in use.

On the brink of death, animals treated with *dBfp*-30 or *dBfp*-60 showed typical toxic signs such as muscular weakness and clonicity hyperspasmia. Other clinical signs, including appearance, eating and drinking habits, motor activity, and behavioral

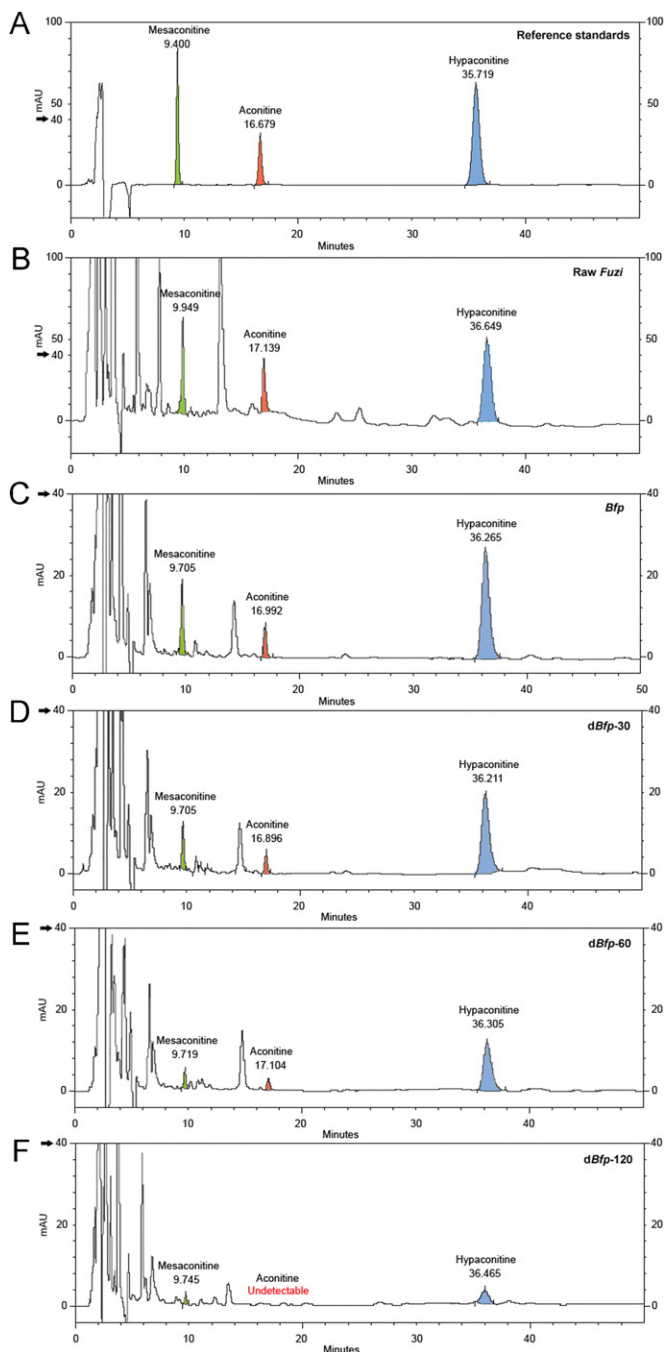


Fig. 2. HPLC chromatograms of aconitine, mesaconitine, and hyaenitine in (A) reference standards mixture; (B) raw *Fuzi*; (C) *Bfp*; (D) *dBfp*-30; (E) *dBfp*-60; and (F) *dBfp*-120. The chromatographic peak of each compound could be well separated with different retention times.

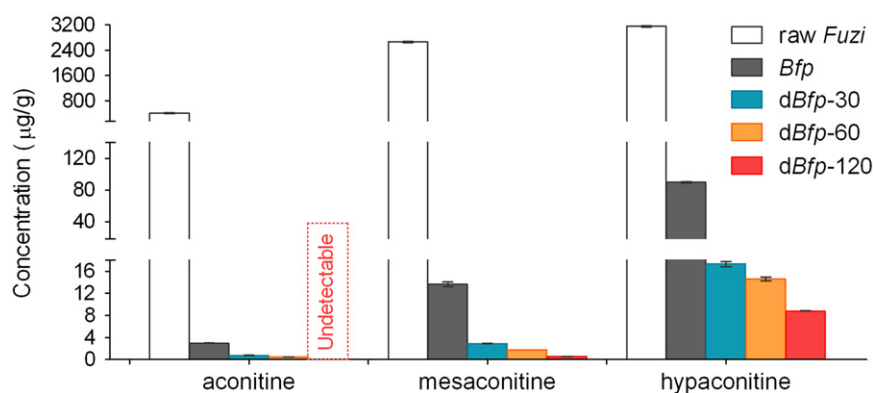


Fig. 3. Concentrations of aconitine, mesaconitine, and hyaconitine in raw Fuzi, Bfp, dBfp-30, dBfp-60, and dBfp-120. Values are means $\pm$ SD, $n=3$.

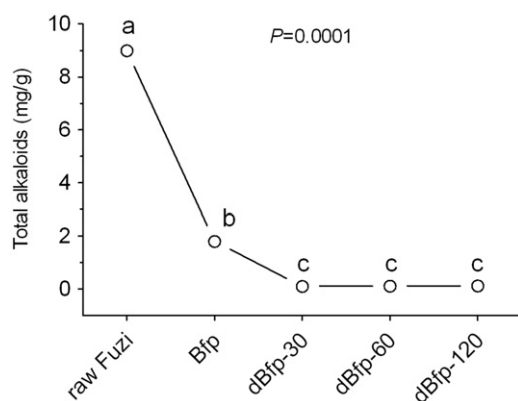


Fig. 4. Amounts of total alkaloids in raw Fuzi, Bfp, dBfp-30, dBfp-60, and dBfp-120. The data of each sample with different lowercase letter on its spot differed very significantly (Fisher's LSD, $P=0.0001$). Values are means $\pm$ SD, $n=3$.

peculiarity, remained normal in each group with whatever treatment used. Postmortem and histopathological examinations revealed no macroscopic or microscopic difference in size, shape, color, and texture of internal vital organs between dead and survived mice. However, the body weight gain of mice varied dramatically among different groups with different treatment regimes, as shown in Table 2. A stepwise decline of body weight gain was displayed with increasing concentrations of each detoxicated Fuzi, especially for dBfp-30 and dBfp-60 (Table 2). As compared with the control group, a slight weigh loss were observed in groups treated with dBfp-60 and dBfp-120, whereas a significant weight loss was observed in dBfp-30 treated groups ($P < 0.05$). Accordingly, no NOAEL of dBfp-30, dBfp-60, and dBfp-120 could be estimated as 70 g/kg, 100 g/kg, and 130 g/kg, respectively.

3.5. Therapeutic effect of detoxicated Fuzi on AA model

3.5.1. Effect on physiological abnormalities

After AA developed for 15 days, the rat appearance, drink intake, behavioral pattern, and rectal temperature in the model group remained within the normal levels, and the animal's urine and feces were in normal color and regular volume. Statistically significant decrease of body weight was observed in the model group compared to the normal group from day 6 to 15 after the AA onset ($P < 0.01$) (Fig. 6). With respective treatment of dBfp-30, dBfp-60, and dBfp-120, the body weight was significantly recovered from day 6 to 15 ($P < 0.01$), as compared to the model group (Fig. 6). No significant difference was found among the three treatments ($P > 0.05$). There was also a significant change of food intake between the normal group and control group ($P < 0.01$)

(Fig. 6). However, no treatment resulted in recovery of this parameter with respect to the model group ($P > 0.05$), except only dBfp-30 and dBfp-60 restored it toward but not up to normal on day 15 (Fig. 6).

3.5.2. Effect on clinical sign of AA

HPV as the clinical sign of arthritis reflects both inflammatory and arthritic disorders occurring in AA rats. It was dramatically increased by the arthritis process in the model group, reaching significantly higher level than that in the normal group on day 1 and was elevated and maintained above 100% throughout the 15-day period (all $P < 0.01$) (Fig. 7). A recovery effect was observed for dBfp-30, dBfp-60, and dBfp-120 treatments in the treated groups, by which HPV was significantly decreased on day 1 and from day 6 onward (all $P < 0.01$) (Fig. 7). Of importance, the HPV lowering effect enhanced progressively with increasing treatment days from 6th to 15th day (Fig. 7). However, no significant difference of effect was found among dBfp-30, dBfp-60, and dBfp-120 ($P > 0.05$), except on day 1 and day 15 when a slightly better effect was offered by dBfp-30 (Fig. 7).

3.5.3. Effect on immunological parameters in AA

IL-1 and TNF- α , the proinflammatory cytokines, are important immunological factors in contributing to the development and progression of human RA as well as arthritis in animal models. As shown in Fig. 8, the serum levels of both IL-1 and TNF- α in the model group exhibited significantly higher than that of the normal rats ($P < 0.01$), but, with daily administration of dBfp-30, dBfp-60, and dBfp-120, the abnormally higher levels of these inflammatory markers became suppressed significantly and finally attained the normal range. More importantly, such anti-inflammatory effect displayed no significant differences among the three different treatments ($P > 0.05$).

4. Discussion

Traditionally, in China and other Asian countries, the use of TCM herbs is commonplace for the treatment of systematic diseases such as RA. With the increasing popularity of TCM, it is necessary to assure the appropriate use of the herbal medicines in clinic since serious side effects can be caused by toxic herbs (Abbott, 2005). As a typical toxic TCM, Fuzi has attracted much attention and generated great interest for its dual effects, viz., medicinal efficacy as well as toxic property (Liou et al., 2006; Lin et al., 2007; Chen et al., 2009; Lin et al., 2010; Tang et al., 2012). The main active constituents in this herb are a series of alkaloids which can exert analgesic and anti-inflammatory effects in the treatment of RA (Zwerina et al., 2004; Perera et al., 2010).

Table 2
Body weight^a and survival of mice after oral administration of dBfp-30, dBfp-60, and dBfp-120.

Body weight gain (g) at:	dBfp-30 (g/kg)					dBfp-60 (g/kg)					dBfp-120 (g/kg)				
	70	100	130	160	190	70	100	130	160	190	70	100	130	160	190
24 h	1.1	0.4	−0.9	−1.4	–	1.3	0.6	0.0	−0.4	−1.2	1.0	0.8	0.4	0.0	0.1
72 h	4.1	2.8	2.3	0.7	–	4.7	4.5	4.0	0.8	−0.5	4.7	4.2	4.2	2.5	1.9
336 h	10.8	10.9	9.8	7.4	–	11.5	10.9	11.7	10.7	9.7	11.3	11.2	11.5	10.5	10.5
Body weight change at 14 days relative to control	0.1	0.2	−0.8	−4.2	–	0.0	0.4	−0.3	−1.3	−1.3	0.2	0.0	0.1	−0.5	−0.9
Survival rate (%)	100	87.5	75	50	0	100	100	100	100	62.5	100	100	100	100	100

No value available due to death.
^a Values are means of replicates in each group.

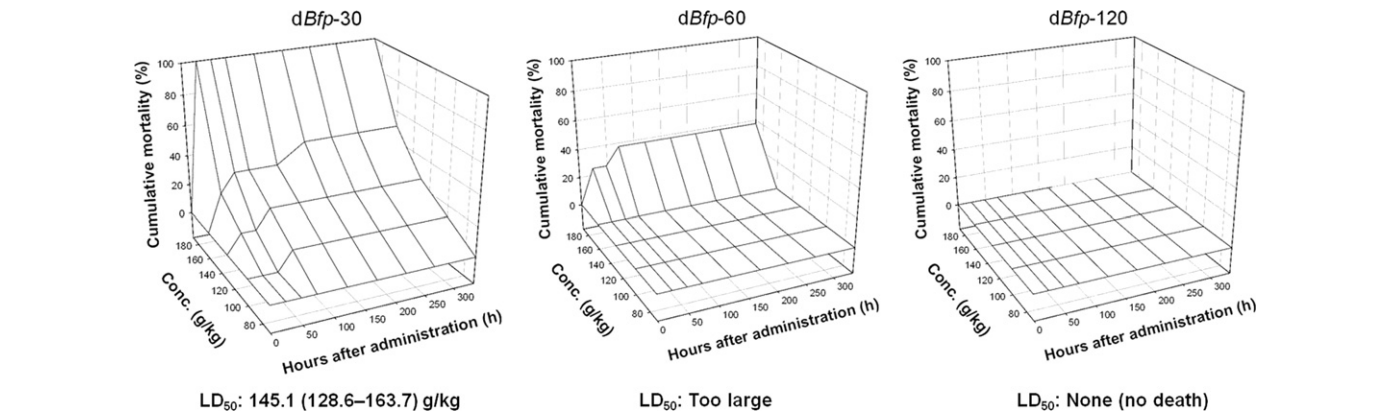


Fig. 5. Cumulative mortalities of mice after oral administration of dBfp-30, dBfp-60, and dBfp-120 (g/kg) with LD₅₀ value and associated 95% confidence limits.

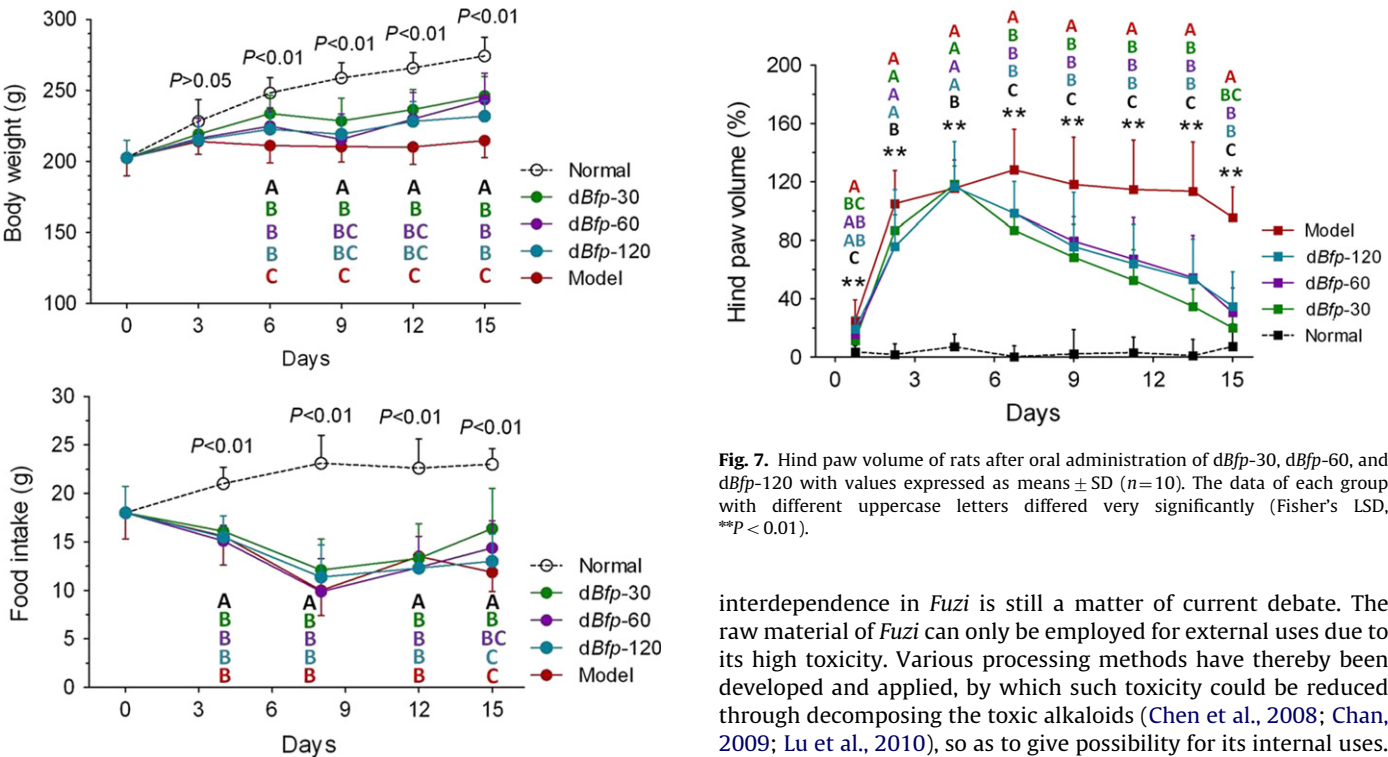


Fig. 6. Body weight and food intake of rats after oral administration of dBfp-30, dBfp-60, and dBfp-120 with values expressed as means ± SD (n=10). The data of each group with different uppercase letters differed very significantly (Fisher's LSD, $P < 0.01$).

Of these, however, aconitine, mesaconitine, and hyaconitine are considered to be the major toxic ingredients (Wong et al., 2002; Liou et al., 2005). Thus whether toxicity and efficacy are of

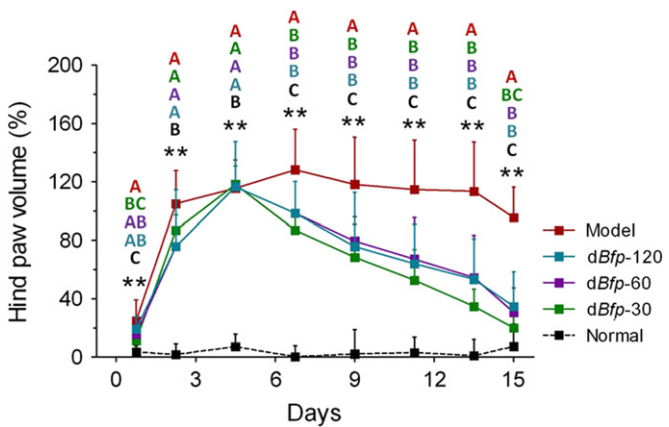


Fig. 7. Hind paw volume of rats after oral administration of dBfp-30, dBfp-60, and dBfp-120 with values expressed as means ± SD (n=10). The data of each group with different uppercase letters differed very significantly (Fisher's LSD, $^{***}P < 0.01$).

interdependence in *Fuzi* is still a matter of current debate. The raw material of *Fuzi* can only be employed for external uses due to its high toxicity. Various processing methods have thereby been developed and applied, by which such toxicity could be reduced through decomposing the toxic alkaloids (Chen et al., 2008; Chan, 2009; Lu et al., 2010), so as to give possibility for its internal uses. However, to date, patients who need to intake *Fuzi* remain at high risk of poisoning, since the available processing and depoisoning methods are both inadequate for use (Tai et al., 1992; Chan et al., 1994; Chan, 2009). Even if the toxicity in *Fuzi* could be completely eliminated, how can we ensure its efficacy still existing there-after? To fill the above gaps in knowledge, the current study was conducted in three parts: Part (a), Part (b) and Part (c), as mentioned in Section 1.

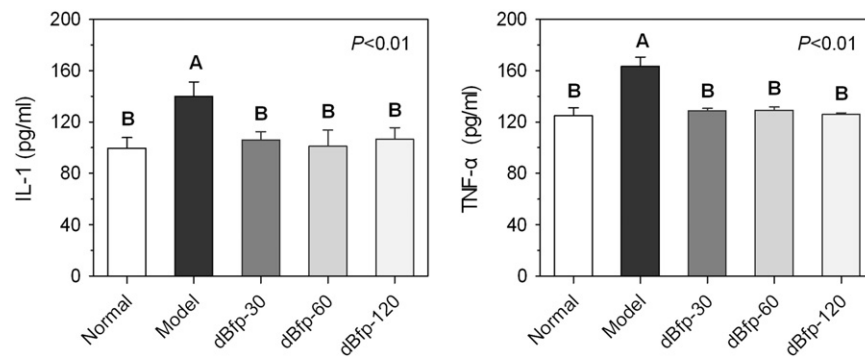


Fig. 8. Rat serum levels of IL-1 and TNF- α after oral administration of dBfp-30, dBfp-60, and dBfp-120 with values expressed as means $\pm$ SD ($n=10$). The data of each group with different uppercase letters differed very significantly (Fisher's LSD, $P < 0.01$).

A new detoxication strategy was developed and validated in Part (a), based on the rationale of hydrolysis of toxic alkaloids to its less toxic or non-toxic derivatives (Chan, 2009). Our preliminary study has screened many processed *Fuzi* and found *Bfp* as the optimal one with minimum levels of toxic alkaloids (data not shown). However, it still remains aconite toxicity with LD₅₀ of 26.3 g/kg (CHCSATCM, 1998). Therefore, in this study, we modified *Bfp*'s processing method by adding an extra decoction step to exert detoxication effect. Our data demonstrated that the longtime-decoction could substantially reduce the *Bfp* toxicity at a time-dependent manner, resulting in a complete detoxication of *Fuzi* after 120 min decoction (dBfp-120) at the dose range 70–190 g/kg (Fig. 5). To our surprise, dBfp-120 at 190 g/kg enabled its safe dose to reach considerably high level (about 1200 g) for human. It is very much high than the human lethal dose of the routinely processed *Fuzi* (10–16 g) as well as the wild plant (1 g) (Singh et al., 1986; But et al., 1994). According to the international classifications for hazardous substances for acute oral toxicity testing (OECD, GHS, EPA, and so on), 2–5 g/kg has been designated as the limit dose (upper limitation on testing) beyond which a single oral dose of the tested substance can be considered non-toxic (OECD, 2001a, 2001b; Schleder et al., 2005). In addition, most acute toxicity studies have defined the non-toxicity of plant product as its LD₅₀ is higher than 3–5 g/kg (Horn, 1956; Ogwal-Okeng et al., 2003; Dêciga-Campos et al., 2007; Qin et al., 2009). Therefore, comparatively, dBfp-120 can be regarded as the completely detoxicated *Fuzi* with certain safety after the new strategy treated, due to its no lethality (no morbidity) and very high NOAEL (130 g/kg).

On the other hand, in parallel, we chemically analyzed raw *Fuzi*, *Bfp*, dBfp-30, dBfp-60, and dBfp-120 using a validated HPLC and titrimetric methods in Part (b). Both chromatographic and quantitative analyses have demonstrated a decline of toxic alkaloid levels with increasing decoction times and even found an elimination of aconitine in dBfp-120 (Figs. 2 and 3). This finding explained why *Bfp* toxicity could be reduced with increasing decoction time. Considering the rapid onset of life-threatening aconitine poisoning as well as the little possibility of eliminating circulating aconitine in vivo (Tai et al., 1992; Lin et al., 2004), the highly toxic aconitine (645.7 kDa), with lipid solubility and lethal dose of 2 mg/kg for human (LD₅₀ of 1.8 mg/kg for mice), seems to be the foremost hazardous alkaloid. Thus, it is reasonable to infer that the complete detoxication of dBfp-120 is mainly ascribed to the elimination of aconitine and partly to the diminishment of other toxic alkaloids. Further, the small amount of mesaconitine and hypaconitine rested in dBfp-120 might be the factor which has contributed to the adverse effect of dBfp-120 on the weight loss of mice (Table 2). It is well known that the alkaloids in *Aconitum carmichaeli* roots are of analgesic, diuretic,

anti-inflammatory, and cardiogenic activities on which the efficacy of *Fuzi* depends (Suzuki et al., 1994; Ameri, 1998b; Desai et al., 1998; Taki et al., 2003). Such importance of the alkaloids prompted us to measure the total alkaloid amounts in our different *Fuzi* samples for monitoring the alkaloid-content change during the detoxication treatment. Our data found twice dramatic reductions of total alkaloids after the first processing and the 30-min decoction (Fig. 4). In contrast, no difference among dBfp-30, dBfp-60, and dBfp-120 was found ($P > 0.05$), due to either the no influence of decoction time on the alkaloid contents or that the residual amount of alkaloids was too small to identify the difference. This result raises the question of whether the detoxicated *Fuzi* with such small total alkaloid amount remains effective in treating diseases such as RA.

To answer this question above, Part (c) conducted a pharmacological experiment using AA model in rats. AA is a specific experimental model for RA, by which typical RA symptoms would be caused (Ding et al., 2003). After AA model developed, sustained anorexia and body weight loss occurred in AA rats (Fig. 6), which was in line with the common features of RA (Roubenoff et al., 1994). However, only the body weight could be restored by the detoxicated *Fuzi* (Fig. 6), suggesting no orexigenic effect exerted. A remarkable paw swelling found in AA rats indicated the typical onset and development of our AA model. To some extent, this clinical sign reflects the severity degree of inflammation, thereby implying a severe inflammatory response in the joint of the AA rats. Therefore, we measured the important proinflammatory cytokines, IL-1 and TNF- α , to evaluate the immunological abnormality in AA as well as the anti-inflammatory effect of detoxicated *Fuzi*. IL-1 and TNF- α have been reported to play pivotal role in the pathogenesis and pathophysiology of RA (Dinarello, 2002), which synergistically mediate synovial inflammation, pannus formation, cartilage destruction, and bone erosion by stimulating the synoviocytes, chondrocytes, and osteoblasts (Choy et al., 2002; Dinarello and Moldawer, 2002). Our data have demonstrated such inflammation occurred in AA rats and revealed the anti-inflammatory effect exerted by all detoxicated *Fuzi* (Fig. 8). The inhibition of IL-1 and TNF- α could promote a great progress in the therapy of RA (Bauerová et al., 2011). Interestingly, however, no matter what parameters we tested in this pharmacological experiment, no significant difference could be seen among dBfp-30, dBfp-60, and dBfp-120 for their therapeutic effects. This finding reveals that the different time lengths of decoction (30, 60, and 120 min) have little influence on the efficacy of detoxicated *Fuzi*, further indicating that dBfp-120 with both no toxicity and anti-AA effect is the best one and its detoxication strategy is most optimal. Thereby we can conclude that *Fuzi* is not of toxicity–efficacy interdependence and its safe and effective application in clinic, at least for RA therapy, is possible and expectable.

Up to now, except ours, there is no method available for complete detoxication of *Fuzi*, and as yet no literature exists regarding its toxicity–efficacy relationship. Henceforth, *Fuzi* can be completely detoxicated and be safe and effective used in clinic. These findings have supported our hypothesis that *Fuzi*'s toxicity is not essential for its therapeutic efficacy especially against AA. Here, accordingly, we define that the complete-detoxication strategy must comprise *Bfp*-applied processing and 120-min decoction under our experimental condition, and an ideal detoxicated *Fuzi* should meet the following preconditions: (i) no mortality or intoxication sign in acute toxicity test; (ii) no detectable aconitine and very little residual of other toxic alkaloids inside; and (iii) retention of a therapeutic efficacy. Our methodology is reasonable and acceptable for supporting the findings in this study. For instance, only employing acute toxicity test was appropriate and enough for the toxicity evaluation of *Fuzi*, since the onset of aconitine poisoning is relatively quick and occurs within 10 min to an hour post-ingestion due to the rapid absorption of toxic alkaloids by the upper gastrointestinal tract (Lin et al., 2004). Pharmacokinetic study has also supported this fact (Wang et al., 2005; Tang et al., 2012). Moreover, it is desirable to only analyze aconitine, mesaconitine, and hypaconitine for explaining the detoxication mechanism, because these three alkaloids are not only the major toxic ingredients but also the active compounds acted as the specific markers in the raw *Fuzi*, processed *Fuzi*, and *Fuzi*-contained TCM preparations (Wong et al., 2002; Chen et al., 2008). Further, the dose range of detoxicated *Fuzi* (70–190 g/kg) in the acute toxicity test was chosen from preliminary experiments to maximally determine the possible safe dose range, whilst the high dose of *dBfp*-120 (14 g/kg) used in the pharmacological experiment was derived from the high human dose (120 g) recommended by TCM clinical doctors (Sun, 2007; Yang, 2008). On the whole, this study has established a standard of complete-detoxication strategy for *Fuzi* and thereby supported its safe and effective application. The implication is that this standardized and validated strategy may be useful for the detoxication for other toxic analogs. More importantly, the non-interdependent toxicity–efficacy relationship of *Fuzi* can be well explained by the fact that the non-toxic aconine, mesaconine, and hypaconine in *Fuzi*, derived from the toxic DDAs during detoxication, also have potent medicinal effects (such as cardiac, analgesic, and immunological activities) (Suzuki et al., 1993; Kobayashi et al., 1998; Liu et al., 2012). Further, our findings may provide new clues and guidance for the quality control criteria of *Fuzi*, since the current one adopted by the China Pharmacopeia is limited with respect to test only for aconitine (China Pharmacopeia Committee, 2010). Regarding the ineffectiveness of RA treatment in many cases, the new obtained *dBfp*-120 can further be developed as a promising candidate of herbal drug for RA therapy. The next step in future will be to determine the chemical composition and other efficacy of the completely detoxicated *Fuzi*, *dBfp*-120.

5. Conclusion

An optimized detoxication strategy, comprising *Bfp*-applied traditional processing and long-time (120 min) decoction, has been originally and successfully established in this study for the complete detoxication of *Fuzi*. The essence of the detoxication mechanism is the hydrolysis of toxic alkaloids. *dBfp*-120 has been screened out as the ideal detoxicated *Fuzi* with both non-toxicity and therapeutic efficacy, verifying the non-interdependence of *Fuzi*'s toxicity–efficacy relationship as well as our hypothesis. This is the first report of such a detoxication strategy available for *Fuzi* and its application in AA treatment. In sum, our findings have

provided a detoxication standard and contributed to a better understanding of toxicological and pharmacological profiles for *Fuzi*. It will further offer a new clue and idea for detoxicating other toxic herbs and benefit the development of safe and effective herbal drugs for disease treatment, especially for RA therapy.

Acknowledgments

This study was supported by the Key Project of Zhejiang Provincial Administration of Traditional Chinese Medicine of China (Grant no.: 2008ZA004) and the Zhejiang Provincial Natural Science Foundation of China (Grant no.: LQ12H27003).

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